

Manual / Product Information incident reserves for BSPs

Requirements and procedures for incident reserve delivery

Preface and reader's guide

For maintaining the real-time power balance of the Netherlands, TenneT uses balancing energy from Frequency Restoration Reserves (FRR). All FRR products are offered to TenneT by market parties in their role as Balancing Service Provider (BSP). A distinction can be made between automatic Frequency Restoration Reserve (aFRR) and manual Frequency Restoration Reserve (mFRR). mFRR products can have direct (mFRRda) and/or scheduled (mFRRsa) activation.

This document focusses on the specific mFRR product that TenneT uses for balancing the system, incident reserve. This is a directly-activated product. There is a balancing capacity and a balancing energy product for incident reserve. This document describes all processes regarding incident reserve.

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Version control

The document has several chapters that describe the specific elements of incident reserves (delivery). To have clarity on what has been modified in each chapter, the version control table describes the changes at the chapter level.

Version	Date	
V 1.0	28-02-2023	Initial version
V 1.1	16-05-2023	Textual addition imbalance adjustment and the adjustment of the figure imbalance adjustment due to incorrect display.
V 1.2	01-06-2023	Clarification of partial activation and adjustment of the delivery of incident reserves in case of transport limitations
V 1.3	01-08-2023	Changed to reflect the incident reserve balancing energy product due to implementation of energy bids for incident reserve (formerly only balancing capacity)
V 1.5	01-11-2024	Change after internal review.
V 1.6	02-01-2025	Changed after review BSP
V 1.7	03-02-2025	Changed after BSP meeting
V 1.8	05-06-2025	Changed for MARI implementation paragraph 2.4, activated energy messaeags and sanctioning
V1.9	03-07-2025	Changed paragraph 7.2 imbalance adjustment based on activated energy information.
V1.91	13-01-2026	Changed on monitoring details, aggregation of reference values and measurement values, activated energy information and imbalance adjustment.

1. Definitions and abbreviations

Balancing service provider	BSP
Balance responsible party	BRP
Imbalance settlement period	ISP
Day prior to the day of execution	D-1
Load Frequency Controller	LFC
Manual FRR directly activated	mFRRda
Automatic FRR	aFRR
Balancing energy gate closure time	BEGCT
Auction Platform for Ancillary Services	APFAS
Full Activation Time	FAT
Volume / price combination for 1 ISP	Bid

Table 1: Definitions and abbreviations

2. Balancing capacity and balancing energy product

There are two relevant products for incident reserve:

- The balancing energy product
- The balancing capacity product

The balancing energy product can be delivered without a contract for balancing capacity. A contract for balancing capacity obligates the BSP to offer a certain volume of balancing energy bids.

The product requirements for the balancing energy and balancing capacity product can be found in the Netcode Elektriciteit respectively. Additional details can be found in this document.

2.1 Summary of the balancing capacity product

TenneT must have a minimum quantity of FRR available. The dimensioning of these minimum quantities is required via European regulations. Further explanation of dimensioning of FRR can be found on the TenneT website¹ and the Netcode Elektriciteit.

To ensure that sufficient FRR capacity is always available, TenneT contracts incident reserves from BSPs.

¹<https://www.tennet.eu/nl/elektriciteitsmarkt/ondersteunende-diensten/>
<https://www.tennet.eu/nl/elektriciteitsmarkt/ondersteunende-diensten/> .

The capacity product Incident reserve is procured asymmetrically (either upward or downward) with a minimum size of 1 MW and step sizes of 1 MW.

The bid requirements are laid down in the Netcode Bijlage 10.

2.2 Summary of the balancing energy product

TenneT needs balancing energy bids to be able to activate incident reserve. These bids are sent by BSPs in accordance with the product requirements from the Netcode until the balancing energy gate closure time (BEGCT), 25 minutes before the validity period of the bid. At the balancing energy gate closure time the bid becomes firm and the BSP is obligated to deliver upon the bid when activated. Bids are only activated for their validity period of 1 ISP as described in chapter 4.

All balancing energy bids are by default considered divisible, however, the BSP can indicate when a bid is indivisible up to a volume of 100MW.

2.3 Time limitations

A BSP that is prequalified for incident reserve product, signs a framework agreement. During the prequalification, a BSP must indicate if these time limitations are needed / used.

2.3.1 Maximum duration

During the prequalification, the BSP can indicate that part of the assets used in the portfolio have a maximum duration when delivering. The delivery of the volume with the time limitation maximum duration must be delivered for 4 ISPs in a 2-hour period as a minimum. After that, a BSP can withdraw its energy bids without penalty for the duration indicated during the prequalification.

The BSP only indicates a maximum duration if this is needed from a technical or business reason which cannot be discounted in the bid price.

2.3.2 Waiting time

During the prequalification, the BSP can indicate that part of the volume of the portfolio has a waiting time between activations. The maximum waiting time is 24 ISPs (6 hours).

The BSP only indicates a waiting time if this is needed from a technical or business reason.

After that, a BSP can withdraw its bids without penalty for the duration indicated during the prequalification.

The BSP only indicates a maximum duration if this is needed from a technical or business reason which cannot be discounted in the bid price.

2.4 Requirements for installations

The installation for the supplied or withdrawn power needs to be connected to the Dutch grid with an allocation point, either a primary or secondary allocation point. These must be allocation points to the Dutch Load Frequency Control area, not necessarily the TenneT grid.

An individual installation can only be activated in one direction at a time. A BSP can only have an upward and downward regulation obligation if it has enough installations connected to different allocation points to guarantee delivery in both directions without simultaneously upwards and downwards delivery at any individual allocation point. Therefore, the total available power of the assets in the BSP's portfolio must be at least equal to the sum of the volume in the send energy bids (up and down).

Regarding the verifiability, no incident reserves can be supplied at an allocation point on which aFRR, (in future) MARI product or any other energy regulation² is also active at the same time without separate metering, except for FCR. See the illustration below which configurations are allowed, and which are not. If the BSP is delivering incident reserves with the same unit as with FCR, then the burden of proof is on the BSP to demonstrate, if requested, that both calls have been met when required. An Exception to delivering aFRR and incident reserve simultaneously at the same allocation point are:

- if incident reserve is activated between t-15 and t-0 minutes for ISP 1 of an incident reserve contracted period and delivering contracted aFRR in the last ISP of the previous contracted period.
- if incident reserve is delivered in the last ISP of a contracted period and the aFRR delivery starts at the first ISP of the next contracted period.

TenneT has the right to perform a compliance check on the data processing (including the reference signal) and relevant procedures (including ability to activate power in both directions) of a BSP by means of an audit, carried out by an independent party.

² This includes schemes other than Automatic Frequency Restoration Reserve (aFRR) or Incident reserve

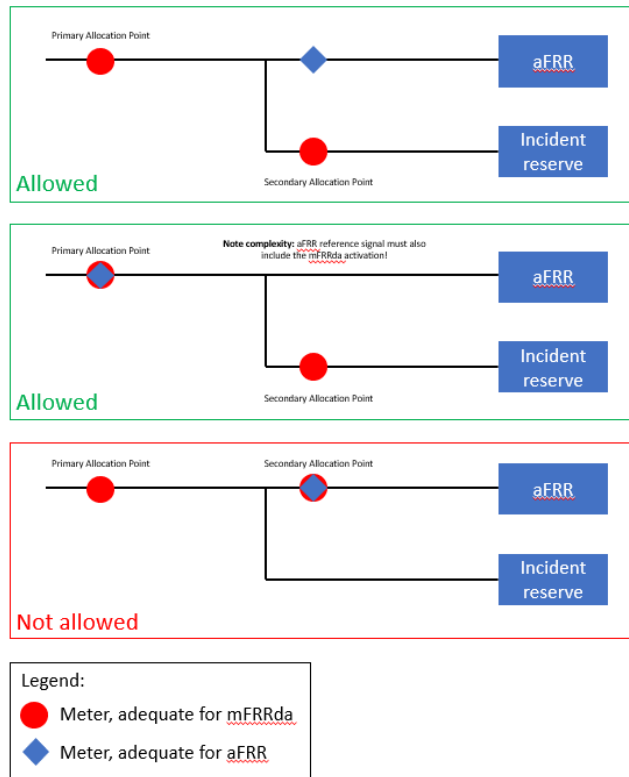


Figure 1: allowed configurations for simultaneous delivery of aFRR and incident reserves

2.5 Delivery of balancing energy from connection points with transmission right limitations

Balancing energy from incident reserve is a portfolio product. If a connected party is subject to a restriction on his transmission rights e.g. because of a capacity restriction contract this needs to be considered in the delivery of incident reserve from the allocation point belonging to the connection point. Such restriction will not lead to compensation of the BSP or exemption of the BSP's obligation to deliver upon firm bids or balancing capacity.

The connected party is responsible for the communication of the restriction of transmission rights to the BSP. In case of a restriction on the transmission rights affecting the ability to deliver from specific allocation points in the BSP portfolio the BSP should ensure that no balancing capacity bids are offered (before the applicable gate closure time) or energy bids are placed (before BEGCT) that depend on such delivery.

When the restriction of the transmission right is imposed after the FRR capacity auction, the BSP is responsible for ensuring the delivery of the product. The BSP had several options for this:

- Ensure the required balancing energy bids can be delivered from other allocation points in its portfolio
- transfer the obligation to another BSP (before d-1 23:00)

If it is not possible for the BSP to guarantee the product delivery they should withdraw its energy bids or not send in energy bids. The consequences will take place as described in paragraph 6.3 on sanctions.

If the BSP can demonstrate that it was not possible to deliver the FRR capacity or energy product in a different way, this will not result in a penalty. A decision on this lies with TenneT.

2.6 Interaction with delivery of other ancillary services

A BSP can provide balancing energy from incident reserve with individual assets or with a pool of assets indicated by a set of allocation points, when prequalified. Concurrent delivery of balancing energy from incident reserve with the delivery of other balancing energy products at the same allocation point(s) is not allowed, except for as described in paragraph 5.1. Delivery of FCR and balancing energy from incident reserve can be concurrent at the same allocation point(s).

Redispatch products can be concurrently delivered from the same allocation points as incident reserve balancing energy. Delivery of balancing energy from incident reserve is possible from allocation points where capacity restriction contracts are active, however, the below applies in relation to transmission right on the relevant connection points.

3. Capacity auction

For procuring incident reserve capacity, TenneT organizes capacity auctions through the Auction Platform for Ancillary Services (APFAS). In APFAS, FRR auctions are organized where capacity bids of aFRR and incident reserves are selected simultaneously. The awarding algorithm considers the minimum required amount of aFRR and the minimum total FRR and then awards the cheapest total FRR package (aFRR and incident reserves).

Upon award, the BSP receives the price of the bid (pay-as-bid). The auction takes place daily on the day prior to the day of execution (D-1) at 9:00³. The contract duration is in principle one calendar day and runs from 0:00 – 0:00. In special situations (e.g. a solar eclipse), TenneT may decide to organize an additional auction for a shorter period. Within this contract period, the BSP has the obligation to send in energy bids

³ UTC + 01:00, Amsterdam time

with a minimum of the contracted volume for the entire contracted period. The full activation time may be outside the validity period of the bid. This means that the BSP can receive the activation 15 minutes before the start of the validity period of the energy bid.

4. Energy bids

Incident reserve is activated based on energy bids. These energy bids can be submitted by a BSP following a contractual obligation or on a voluntary basis.

The requirements of energy bids are laid out in the Netcode Bijlage 20.

It/they must meet the following characteristics:

1. Energy bids have a unique identification per ISP
2. Energy bids are asymmetrical (either for upward or downward regulation)
3. Energy bids have a bid volume between 1MW and 9999MW
4. Energy bids are valid for 1 ISP (15 minutes)
5. Energy bids can be activated from 15 minutes before the ISP until the start of the ISP
6. Energy bids can be deactivated during an ISP
7. Energy bids are divisible by default
 - a. Indivisible bids are allowed but with a maximum volume of 100MW
 - b. Indivisibility of a bid must be indicated by the BSP in the bid
8. Status of the bid:
 - a. New
 - b. Changed
 - c. Withdrawn

If the availability of assets in the BSP portfolio changes after sending of a balancing energy bid, the BSP must change the already sent balancing energy bid. The change can be the volume and or price. If the complete volume of the balancing energy bid is no longer available due to the unavailability of the asset(s) the balancing energy bid needs to be withdrawn before gate closure time of the ISP.

In the prequalification process it is possible to indicate under certain condition that a maximum delivery duration and/or waiting time should apply for certain assets. Energy bids are sent for the volume of the assets for which maximum duration and / or waiting time apply. Separate energy bids are sent so these bids can be withdrawn without affecting the availability of volumes of assets with no time limitations. When assets reach the time limitations the corresponding energy bids are withdrawn before the BEGCT.

5. Activation process

When there is a need for incident reserve TenneT will determine the needed volume. Based on this, the volume energy bids will be activated until the volume is reached. The activation is based upon the merit order list of an ISP. During the entire period in which TenneT needs incident reserve, the activated bids will be selected based on the incident reserve merit order list of every ISP in the period. BSPs will be activated

every ISP if needed. A BSP is not expected to deliver longer than the ISP for which the BSP is activated.

The bids will be selected from the incident reserve merit order list. The incident reserve merit order list consists of all bids accepted before the gate closure time of the ISP. The incident reserve merit order list is ordered as follows:

- price (ascending for upward volume and descending for downward volume)

Based on the requested volume the bids are selected from the merit order list for activation. Bids with equal prices will be activated proportionally to their bid size. Any non-integer bid activation will be rounded to the nearest integer MW bid-setpoint-value. The BSP receives, the sum of its bid-setpoint-values per direction as total incident reserve setpoint per direction. Only bids that (after rounding) lead to a non-zero setpoint are considered in the market clearing price. If any bid is indivisible but not the complete volume of that bid is needed, then this indivisible bid will not be activated. In that case the volume will first be redistributed (proportionally to volume) to divisible bids with the same price and next to bids with higher (lower for downward volumes) prices. This means that indivisible bids can be paradoxically rejected, meaning that a bid is not activated despite its price being within the marginal price.

In case fully activating an indivisible bid would lead to lower costs than partially activating a next divisible bid but leads to overfulfilment of the requested volume, the divisible bid will be selected and not the indivisible bid. It may be that the exact requested volume cannot be fulfilled with the logic before. In that case the TenneT operator is requested to choose between the nearest feasible solutions.

If during an ISP there is a need for additional volume this volume will be activated for the following ISP. In the appendix 1 different activation scenarios are described to clarify. During an activation, the BSP is informed on which of its bids are activated using the bid identification of the BSP.

When sending an activation, the confirmation by the BSP on reception of the activation has to be received by TenneT within 1 minute after sending the activation. If TenneT has not received the confirmation within the specified time period the activation will be marked as not sent. In that case TenneT does not expect the BSP to respond on that activation. TenneT will continue activating the next bid of a different BSP on the merit order list. The BSP is not excluded for possible activations in the following ISPs. A BSP is allowed to send a negative response to an activation based on ex ante agreed time limitations if the BSP was not able to withdraw the energy bids before the BEGCT.

The information exchange point on BSP side must be available for the period where there is an obligation to deliver:

- obligation starts 15 minutes before the start of a contract
- obligation starts 15 minutes before the first ISP where there are energy bids

Unavailability of the information exchange point on the BSP side will lead to a sanction. The availability of the real time connection is periodically checked.

If the real time connection fails after receiving an activation from TenneT the BSP will deactivate at the end of the validity period of the activated bid.

5.1 Contract period changes

In the last ISP before and the first ISP after the start of a new 24-hour contract period it is considered acceptable to deliver aFRR and incident reserve simultaneously on the same allocation point. This because an activation of incident reserve can start at t-15 minutes. This means that the energy bid for the first ISP for incident reserve can be activated in the ISP prior to the start of the contract. For the aFRR delivery it can be that in the first ISP of the incident reserve contract period an aFRR activation is regulated back to 0.

5.2 Withdrawn of bids

An energy bid can be made withdrawn by the BSP or TenneT.

5.2.1 BSP

Bids can be made withdrawn by the BSP if:

1. The maximum delivery period for contracted bids has been reached,
 2. Due to deactivation of bids and assets are in waiting time,
- Bids can be made unavailable until the BEGCT of the ISP for which the bid is valid.

5.2.2 TenneT

Bids will be made unavailable by TenneT if:

- The IT connection with the BSP is unavailable (see section 7.4).

An unavailable bid will not be activated even if the bid has a favourable price.

6. Data Exchange

To provide incident reserves with voluntary energy bids, the following messaging must be set up in accordance with the implementation guides. See Figure 2 for the overview of messaging between the various market participants.

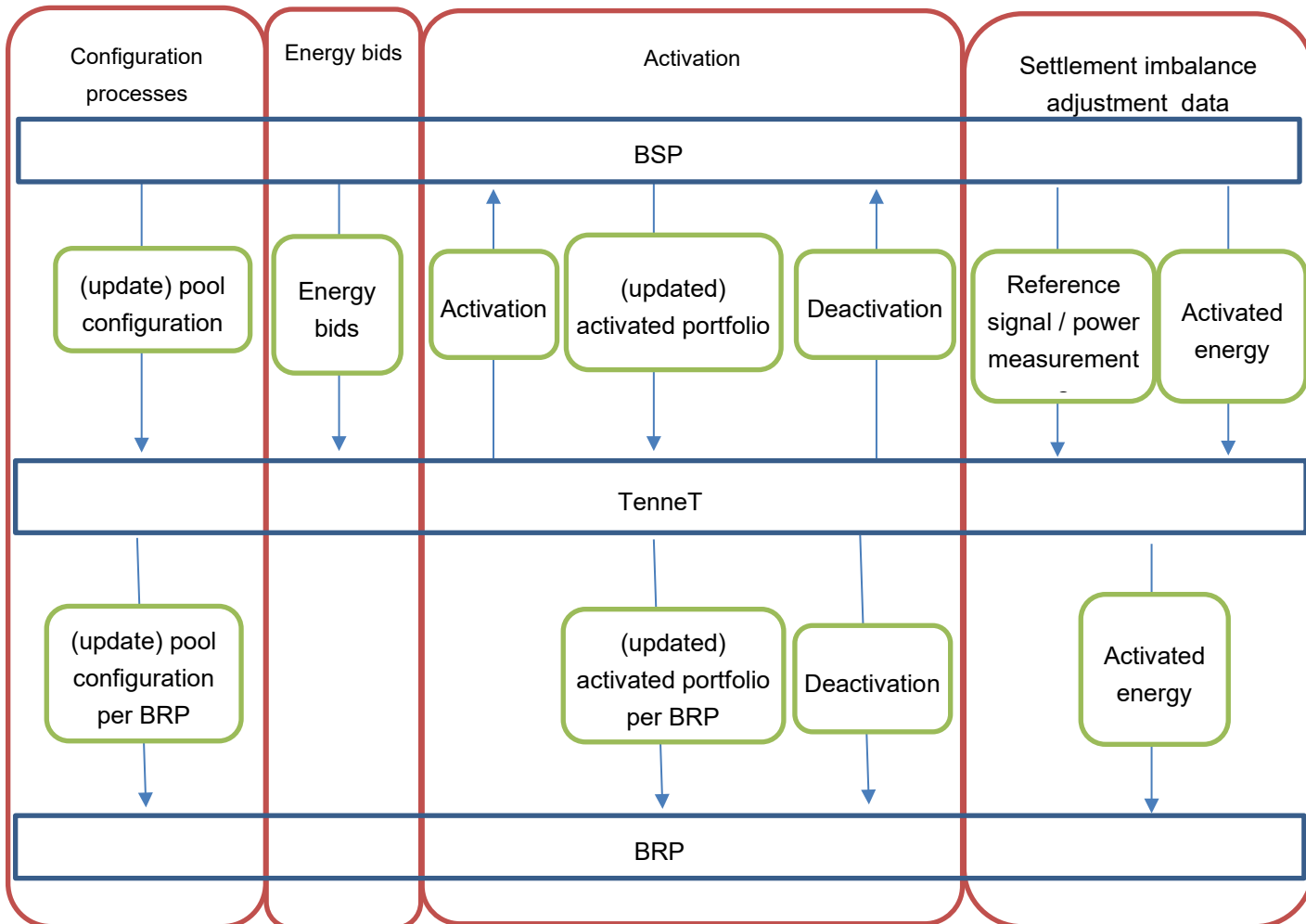


Figure 2 Overview of message exchanges between market participants

6.1 Pool configuration (Configuration process)

The purpose of the pool configuration is to indicate to TenneT which allocation points are active in the BSP's portfolio and which BRP is active at the various allocation points. TenneT notifies the BRP once a month of all allocation points active in an incident reserves pool that are attributed to him. The BSP of incident reserves sends the pool configuration to the at TenneT prescribed manner according to the implementation guide incident reserves activation. This message should be sent monthly, no later than 5 working days before the start of the month

In case of interim changes, the BSP sends an update with an effective date at least 2 days in the future. The last by TenneT accepted message is considered the correct composition of the portfolio of the BSP. A message always includes a complete set of allocation points active in the portfolio of the BSP during the assigned period.

6.2 Data exchange on demand (activated portfolio)

The offered power is called for activation by TenneT by means of a message. This message is sent to the BSP, so that it knows how much power to supply. The BSP informs TenneT, within 5 minutes (this is the deadline), on which allocation point(s) it has activated the called power and with an indication of the quantity. The indicated quantity may not exceed the prequalified capacity. TenneT then immediately informs the BRPs of the allocation point(s) that have been assigned by the BSP with the indication of the amount of power this has activated so that the BRP does not perform a counteraction.

If the BSP fails to send an activated portfolio message with the correct used allocation points and power volumes within 5 minutes after the sending of the activation, the activated portfolio message can be updated until 2 working days after the execution day. An update after 5 minutes after sending of the activation will lead to a sanction (see paragraph 7.3.3).

Every activation is an individual activation containing to total requested volume of all activated bids. If during an ISP a second activation is sent by TenneT the BSP has to deliver the new requested volume. The second activation will contain the sum of the already activated volume and the newly activated volume. In this case, the BSP only must distribute the newly activated volume (difference between new and previous activation) over allocation points. The activated portfolio message based on the new activation contains all allocation points with which the total volume is delivered.

As the allocation points are used to check the activated energy messages the activated portfolio messages can be updated until D+2 (working days)

6.3 Measured values

The incident reserves BSP must facilitate TenneT in such way that TenneT is able to verify that the power adjustment requested by TenneT is realized correctly. TenneT places the following requirements on the verification options:

- Measurements are made by a licensed metered data responsible party
- The BSP receives the measurements in 5-minute kWh value from the metering responsible party or

generates the 5-minute value based on the measurements made by a licensed metered data responsible party using a so called P1 port.

- The BSP sends the measurements to TenneT with 5-minute measurement period for at least the activated period with the last 5-minute measurement period prior to the start the activation.

Chapter 8 describes how the measurements are used to verify the delivery and the determination of the volume of delivered energy.

6.4 Activated energy

In the activated energy message, the BSP indicates the, by the BSP, requested energy at the allocation points. The sum of the requested energy at the allocation points is equal to the energy TenneT has requested from the BSP.

The following rules apply:

- Only allocation points which are present in the activated portfolio message can be used.
- The requested energy is maximum the calculated energy based instantaneous delivery of the prequalified power of the allocation points, the minimum value is 0 kWh .
- The sum of the activated energy is equal to the requested energy by TenneT.
- The first activated energy messages have to be received by TenneT on D+1 4:00 hour.
- Activated energy messages received between D+1 04:00 hrs and D+2 working days are subject to sanctions
- Updates on activated energy are accepted and processed until D+2 Working days.
- Activated energy message received after D+2 working days will be rejected.

Example

If TenneT requests an activation of 30 MW for 15 minutes from a BSP. The BSP divides the requested volume (send in the activated portfolio message) as follows:

- Allocation point A 10 MW
- Allocation point B 20 MW

The allocation points have a prequalified capacity of:

- Allocation point A 40 MW
- Allocation point B 80 MW

The calculated maximum energy based on the prequalified capacity:

- Allocation point A 10 MWh
- Allocation point B 20 MWh

The activated energy message may contain:

Allocation point	Minimum	Maximum
A	0 MWh	10 MWh

B	0 MWh	20 MWh
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The requested energy by TenneT is 7,5 MWh (30 MW for 15 minutes). The sum of the activated energy from allocation points A and B must be 7,5 MWh. The energy per allocation point must be between the minimum and maximum values in the table above.

6.5 Data connection monitoring

TenneT monitors whether the connection for the activation signal to the BSP is operational. If the connection is not operational, TenneT will mark the balancing energy bids as technically unavailable and will not include these balancing energy bids in the merit order list.

The BSP has to monitor the real time connection itself to know whether the connection is operational.

6.6 Capacity product

To provide the incident reserve capacity bids the information exchanges are described in APFAS documentation on MyTenneT.

6.7 Data connection testing

When there are changes in the processes or IT. Depending on the changes these will be tested prior to implementing them in operation. For this, TenneT has an IT environment in which these changes can be tested. If the changes need to be tested with BSP's, these tests with the BSP's can be performed without interfering with operations. This requirement is for all IT connections for the ex-ante, real time, and ex-post.

7. Settlement, imbalance adjustment and penalties

The settlement volumes and imbalance adjustment volumes will be based on the requested energy. The imbalance adjustment will be contributed to the ISP's corresponding to the volumes in the rectangle in Figure 3: determining the requested volumes.

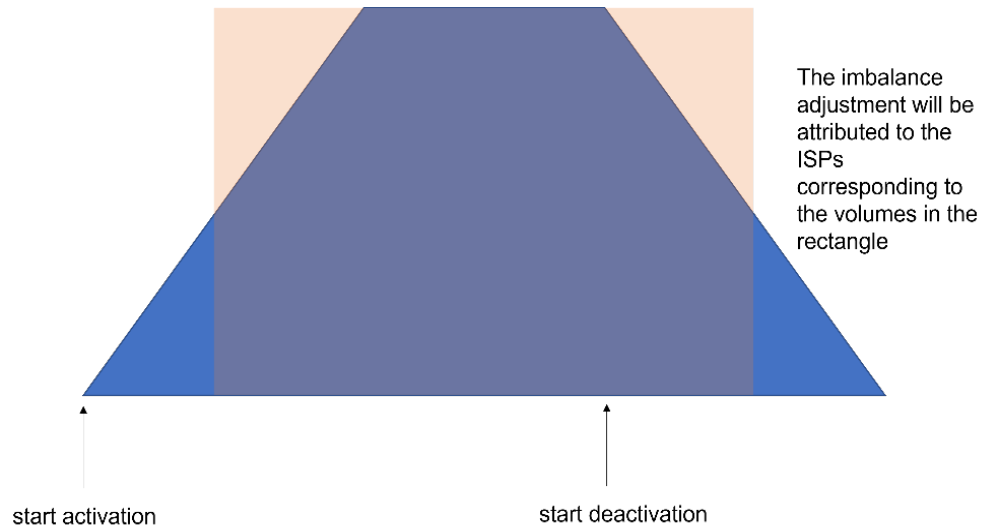


Figure 3: determining the requested volumes

The volume is determined as follows:

- volume during the start of the delivery period (7,5 minutes x requested power) + volume during continuous delivery period (duration x requested power) + volume during the end of the delivery period (7,5 minutes x requested power). See bid 1 in Figure 4: settlement for different bids

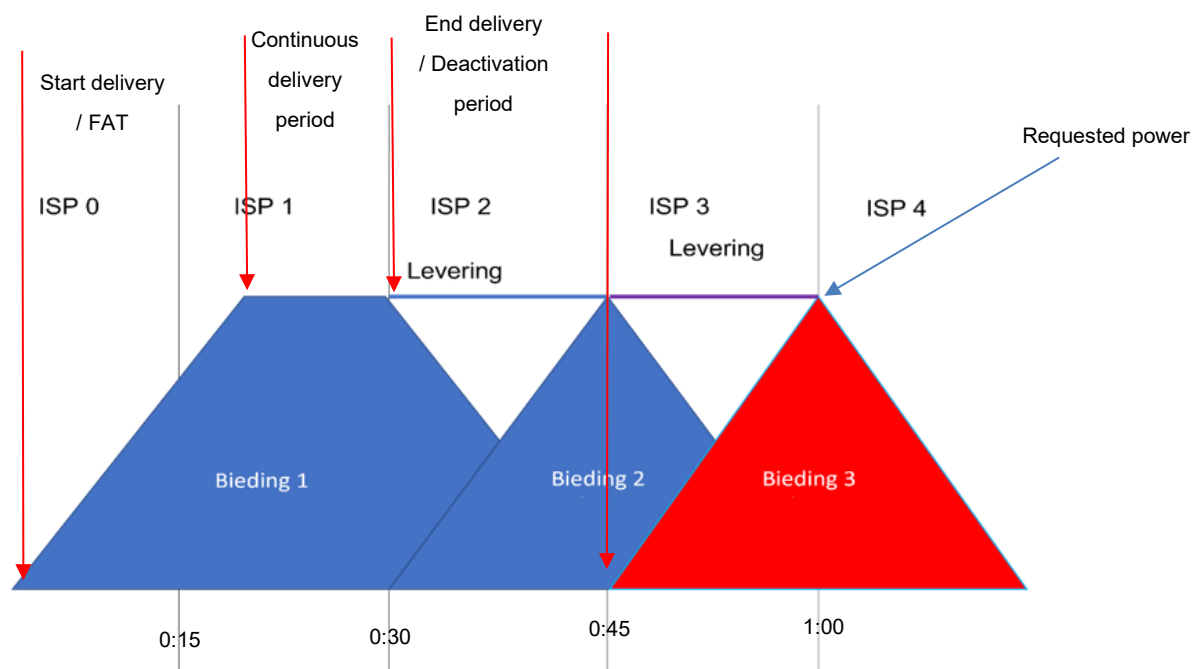


Figure 4: settlement for different bids

Example 1:

Activation bid Bidding 1 (figure 4):

Activation at 00:05 hours

Deactivation at 00:30 hours

Activated power 120MW

Settlement volumes		
50 MWh		
Imbalance adjustment volumes		
00:00-00:15	00:15-00:30	00:30-00:45
5 MWh	30 MWh	15 MWh

Example 2:

Activation bid Bidding 2 (figure 4):

Activation at 00:30 hours

Deactivation at 00:45 hours

Activated power 120MW

Settlement volumes	
30 MWh	
Imbalance adjustment volumes	
00:30-00:45	00:45-01:00
15 MWh	15MWh

7.1 Settlement

The settlement will occur per activated balancing energy bid.

7.1.1 Pricing incident reserves regulation

The settlement price is determined based on the imbalance price methodology. [OBJ]

7.1.2 Determining Remuneration

The remuneration of the BSP is calculated as:

the determined volume (paragraph 8) x the determined balancing energy price (paragraph 7.1.1).

7.1.2.1 Determining remuneration early deactivation

If TenneT no longer needs incident reserve, the activation will be stopped. This will lead to deactivation of the balancing energy bids during an ISP. When an asset used for delivering incident reserve is deactivated during the FAT by request of TenneT the settlement for this asset of the BSP consists of 2 parts:

- for the period, the BSP has been activated based on requested energy based on the activated power of the asset and the balancing energy price (see paragraph 7.1.1).
- for the period, the ISP has not been activated the not requested energy based on the activated power of the asset and the bid price

When a BSP is activated for 2 ISPs and is deactivated 5 minutes into the second ISP. After every activation the BSP sends for every asset used to deliver the activated incident reserve volume the activated power per asset to TenneT. In case of 2 continuous ISP activations the BSP can either deliver both ISPs with the same assets or (partly) change the delivering assets in ISP 2. When a deactivation before the end of the ISP is send in the second ISP the asset(s) which started delivering at the start of the second ISP are deactivated before the end of the FAT. The BSP will be remunerated for the not requested energy of those assets.

Example:

Activation on T-0 in ISP 1 of 100 MW

BSP activates 100 MW on the following assets:

- Asset A for 20 MW
- Asset B for 50 MW
- Asset C for 30 MW

Activation on T-0 in ISP 2 of 100MW

BSP activates 100 MW on the following assets:

- Asset A for 20 MW
- Asset B for 50 MW
- Asset C for 20 MW
- Asset D for 10 MW

After 5 minutes into ISP 2 the BSP is deactivated.

Assets A, B and C have been activated in ISP 1 and 2. As the delivery on these assets is longer than the FAT the energy requested on these assets is remunerated.

Asset D is activated at T-0 of ISP 2. As the deactivation is received in the 5th minute of the ISP asset D is deactivated during the FAT. For asset D the BSP will be compensated for not requested energy.

In this example if asset D would have delivered the whole FAT the requested energy would have been $10 \text{ MW} / 4 = 2,5 \text{ MWh}$. However, the asset is deactivated 5 minutes into the FAT. That means that based on a linear ramp and the requested power (by the BSP to the asset) the reached power is $(10 \text{ MW} / 15) \times 5 = 3,33 \text{ MW}$. Therefore the requested energy is $3,33 \text{ MW} / 60 \times 5 = 0,278 \text{ MWh}$. For asset D the BSP is remunerated for 0,278 MWh with the balancing energy marginal price and for the not requested energy: $2,5 \text{ MWh} - 0,278 \text{ MWh} = 2,222 \text{ MWh}$ with the bid price.

7.2 Imbalance adjustment

For every Imbalance Settlement Period (ISP), starting from the moment a BSP has been activated until the BSP has ramped down after receiving the deactivation of the requested energy will be adjusted by TenneT on the imbalance of the balance responsible party (BRP) of the connection point on which the Incident reserve is delivered (see the figure and examples in paragraph 7). This means that if too little or too much is supplied or withdrawn, this will be settled via the imbalance settlement described in the Netcode elektriciteit.

Each allocation point supplying incident reserves must be accommodated with an approved BRP. This information must be made available to TenneT via the pool configuration message in the manner prescribed by TenneT in accordance with (§ 6.1). The imbalance adjustment is made on the basis of the activated energy message which the BSP sends to TenneT in the manner prescribed by TenneT in accordance with (§6.2). TenneT informs the BRP about imbalance adjustments on its allocation points done based on the activated energy message.

7.3 Sanctions

7.3.1 Not fulfilling capacity obligation

When a BSP is contracted for a certain volume, there is an obligation to send in balancing energy bids for every ISP of the contracted period for the contracted volume. For each ISP where the BSP does not fulfil this obligation the BSP will receive a penalty from TenneT.

Not fulfilling the capacity obligation is determined based on the volume available for activation at the start of the ISP. Unavailability can be caused by not sending in enough volume in the energy bids or when at the start of the ISP bids are excluded from the merit order list due to IT connection failure of the real time IT

connection (activation connection). Exception is when bids are withdrawn based on time limitations.

7.3.2 No or lack of response

The BSP will receive a penalty in case of lack of response when:

- there is no or a negative response to an activation signal from TenneT. Exception is when a negative response is given due to time limitations.
- does not respond with the full power to be determined of course with the acceptable response as described in paragraph 7.3.2.1

As long as delivering aFRR and incident reserve simultaneously at the same allocation point is not allowed the delivery of aFRR should be ended 5 minutes before the potential start of the incident reserve delivery. Exception is the ISP before and after contract period changes. There delivering aFRR and incident reserve is allowed, and lack of response will not be penalized.

7.3.2.1 Determining lack of response

The lack of response is determined during the validation of delivery. The delivery is validated by comparing the reference value to the actual measurement of the allocation point. The reference value is set at the energy volume that is measured at the allocation point(s) with which incident reserves is supplied, in the 5-minute period prior to the 5-minute period in which the activation is send. Because incident reserve is delivered in a portfolio the measurement values and reference values are aggregated into an aggregated reference value (sum of the reference values of the used allocation points) and an aggregated measurement value (sum of the measurement values of the used allocation points). The difference between the aggregated measured value and the aggregated reference value must be equal to the expected response.

A good quality delivery is defined as follows:

- After activation
 - First 5 minutes: No response required, but it is allowed
 - Second 5 minutes: A visible (see table below) response is required unless the full requested power has been reached in the first 5 minutes
 - Third 5 minutes: An additional visible response is required unless the full requested power has been reached in the first or second 5 minutes
- Response required in following 5 minutes is a at least requested energy response from a measurement value in the monitoring
- Delivered energy is at least 90%⁴ of the requested energy this from the start of the ramp up period, the continuous delivery period, the ramp down period.
- After deactivation, the same but opposite conditions as for the activation apply

⁴ This percentage may change over time.

A good quality delivery of a 1 MW activation is defined per minute:

Minute	Delivery	Cumulated delivery
1	0 MWh	0 MWh
2	0 MWh	0 MWh
3	0 MWh	0 MWh
4	0 MWh	0 MWh
5	0 MWh	0 MWh
6	0,001 MWh	0,001 MWh
7	0,001 + increasing	0,002 + increasing
8	0,001 + the increase of the previous minute and additional increase	0,003 + the increase of the previous minute and additional increase
9	0,001 + the increase of the previous minute and additional increase	0,004 + the increase of the previous minute and additional increase
10	0,001 + the increase of the previous minute and additional increase	0,005 + the increase of the previous minute and additional increase
11	An additional increase compared to the previous minute	
12	An additional increase compared to the previous minute	
13	An additional increase compared to the previous minute	
14	An additional increase compared to the previous minute	
15	An additional increase compared to the previous minute	0,113 MWh

As the response is validated based on 5-minute measurement values where the 5-minute periods are fixed the acceptable response per 5-minute period may differ depending on the activation moment. In appendix 2 there are examples of how the acceptable response will look like in 5-minute measurement periods.

The validation of delivery is done per activation. As the BSP has the possibility to change its portfolio after every activation the connection point with which incident reserve is delivered can change every activation. The reference value is determined at the start of the usage of the allocation point in the delivery. If an allocation point is used continuously the reference value is determined in the 5-minute period prior to the 5-minute period in which the first activation is done.

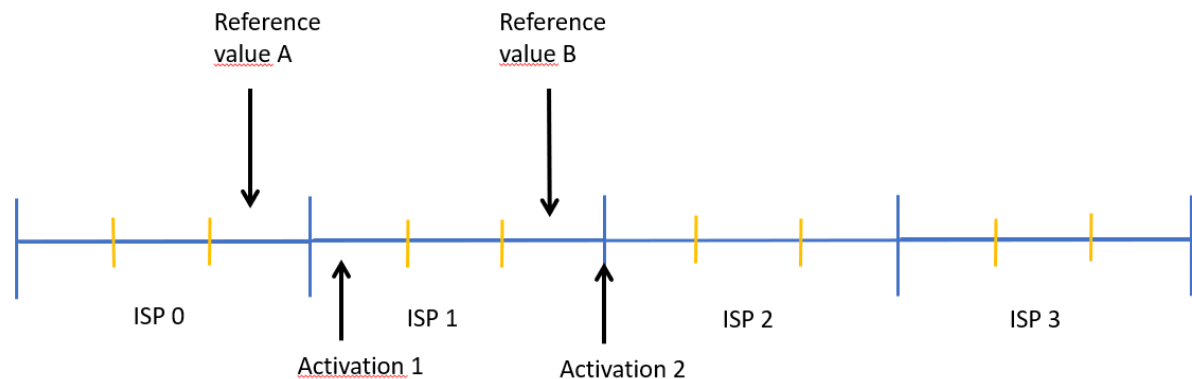


Figure 5: Determining reference value

For the allocation points used in activation 1 the reference value A are used.

For the new allocation points use in activation 2 the reference value B is used.

For the allocation points use in activation 1 and activation 2 the reference value A is used.

7.3.3 Timeliness of administrative processes

If not complying to the administrative deadlines an administrative fee will be issued.

7.3.4 Sanctions

The sanctions will be as follows:

1. Not fulfilling the capacity obligation (no energy bids with the contracted volume available for the full contracted period):
 - a. The remuneration for the not delivered energy is paid back to the payer.
 - b. 10% of the remuneration is paid to TenneT determined based on the absolute from (volume not bid in in an ISP x (capacity price⁵ /4) x sanctioning factor (0,1))
2. Not fulfilling delivery obligation
 - a. The remuneration for the not delivered energy is paid back to the payer.
 - b. 10% of the remuneration is paid to TenneT determined based on the absolute from (volume not delivered in an ISP x settlement price of that ISP x sanctioning factor (0,1))
3. Not fulfilling process obligations in one day leads to an administrative sanction of 250,00 euro with a maximum of 2.500,00 euro per day:
 - a. Manual actions by TenneT in the otherwise automated processes
 - b. Sending information after the initial deadline
4. Availability of the real time information exchange:
 - a. Unavailable at the start of the validity period of the bid when there is a capacity

⁵ The highest price the BSP has received for the capacity based on the initial auction or in case of transfer of obligation to a BSP with no capacity obligation the highest capacity price for the ISP.

- obligation and energy bids have been send in this leads to sanction 1,
b. Unavailable when there is a delivery obligation this leads to sanction 2b

When the capacity obligation or delivery obligation is not fulfilled due to time limitations no sanction will be given.

There will be only 1 sanction for the same ISP. For example if a BSP doesn't fulfil its capacity obligation by not sending in energy bids and for the same ISP the real time information exchange is unavailable this will lead to one sanction. The penalties are relatively limited because BSP will be remunerated for the, by TenneT, total requested volume.

8. Responsibilities for incident reserve in a pool

Forming a pool has the advantage for TenneT of potentially offering incident reserves that otherwise would not be offered. As such, it enlarges the market. A disadvantage, however, is that a pool can complicate administrative and technical procedures for the pool participants, the aggregator (BSP) and TenneT.

A small (production) installation, usually in the case of a pool, usually does not have its own grid connection, but is located behind a grid connection. The BSP, possibly in consultation with his BRP, can choose to have the supply determined on the basis of the measured production or offtake at the (production) installation instead of the measurement at the grid connection. This should also be done on the basis of accountable 5-minute kWh measurements. The measurement⁶ must be carried out by an authorized measurement responsible party. All this, of course, in accordance with applicable laws and regulations.

If multiple installations are in operation behind the connection then the control of one plant may not automatically result in an equivalent but opposite control action of another plant.

Further consequences for pool participants are a matter for the aggregator (BSP) and are outside TenneT's area of responsibility.

9. Pre-qualification

Before a BSP may supply incident reserves, the prequalification process must be completed in accordance with the general steps, described in the prequalification procedure manual, published on TenneT's website. This chapter further discusses the specific prequalification tests and requirement that must take place for incident reserves.

In the prequalification process for Incident Reserve, no distinction is made between applications for voluntary energy bids and contracted energy bids. A party is acknowledged as a BSP for both energy and capacity

⁶ Metering and metering device shall comply with the metering code

bids simultaneously, in parallel to the qualification for their IT connections. This means that there is a single (RPG prequalification) test for the energy and the capacity product. Successful prequalification will already allow participation in the capacity auctions without the need for additional prequalification tests.

The BSP is responsible for adhering to the product requirements in this handbook. If, after being acknowledged as BSP for incident reserve, the BSP does not meet the product requirements, the acknowledgment can be terminated as described in paragraph 4.2 of the prequalification process published on the TenneT website⁷.

9.1 Pre-qualification requirements

In order to prequalify for the supply of incident reserves, the applicant must carry out a test activation at its own discretion and in coordination with TenneT. The activation is carried out by the applicant himself. The delivered response must meet the product requirements described in this handbook. Furthermore:

- The applicant completes and submits an application form (must always be the latest published version) with all Technical Installations (TIs) that it wishes to prequalify (one line per TI) within the applicable deadlines. A correctly filled application form must indicate whether the applicant wants to prequalify for both up- and down-regulating power, or a single direction and the follow-up test(s) must reflect that.
 - In case of application for both upward and downward prequalification, the tests per direction do not necessarily need to be done together, but they must be carried out within the reported test window.
 - Testing outside of an announced (and approved) test window will lead to rejection of the resulting report.
 - The tests should at least demonstrate that: Incident reserve energy bids with a FAT as specified in paragraph of this document, can be regulated
 - A visible response after 5 minutes after activation
 - Deactivation meets the same requirements as for activation
 - The steady-state Incident Reserve requested power and requested energy must be fully provided for at least 15 minutes during the test (i.e. the delivery period excluding the 15 min ramp-up and 15 min ramp down times) per direction to be prequalified.
 - The test must start at least 10 minutes before the first activation (preferably, but not compulsory: from a steady state) for facilitation of determining the reference.
- The incident reserve power to be activated with the selected TIs during the prequalification test has a power of at least 1MW and must match the setpoint sent out by the applicant. This means that partial activation of selected TIs is allowed during the prequalification test.
- By default, it is assumed that TIs do not have maximum duration and waiting time limitations. However, if it is applicable, the applicant shall provide these technical and/or functional specifications for each asset that will be used (by the applicant) for the purpose of providing IR.
 - Maximum duration – the (maximum amount of) time for which the asset is available to deliver during a single activation.

⁷ https://tennet-drupal.s3.eu-central-1.amazonaws.com/default/2025-02/20240809_Prekwalificatieproces_NL_0_0.pdf

- Waiting time – the minimum required downtime for the asset between consecutive activations

These will not be tested, but will be checked ex-post, in case of successful prequalification. Subsequent portfolio additions must include this information as well, if applicable.

The applicant must submit a report based on the tests to TenneT describing the test(s) and their results. Both the report and corresponding data must have the structure, prescribed by TenneT, and describe at least the following:

- Date and time of the test.
- EAN and/or names of the assets used as mentioned in the application form.
- Time-synchronous values of the transmitted setpoint, the measured energy per 5 minutes as raw data and in graph form;
- Analyses in which, based on the data, it is illustrated that:
 - A visible response after 5 minutes after activation is achieved
 - The response is in line with the activated volume
- Analysis in which, using the data, it was explained that the power to be prequalified can be activated within the activation time specified for incident reserves.

The BSP must store the original measurement data (including underlying TI data) of the test with a resolution of at least once every 5 minutes and keep it for 5 years⁸, or until re-prequalification. TenneT can request this data for verification and control purposes.

9.2 Framework agreement

All approved IT-qualifications and prequalification's for incident reserve result in a framework agreement which the BSP must sign before becoming operational. The to be signed version will be send to the BSP by TenneT.

A blank version of this document can be found on the TenneT website.

10. (Operational) Communication

TenneT will need to communicate with all qualified BSP for several reasons, i.e.:

- Operational issues
- Changes in the handbook
- Invitations to workshops on current and future developments

For this communication TenneT will use the contact information registered in MyTenneT. It is therefore expected that the BSP keeps the contact information in MyTenneT up to date at all times. A guide on how to

⁸ 5 years since the prequalification is valid for 5 years.

do update contact information can be found in the FAQ of MyTenneT → my organization.

11. Reference and Resource List

See below for other relevant documents and data sources related to incident reserves delivery.

General information about ancillary services:
[Support services \(Netherlands\) \(tennet.eu\)](https://tennet.eu/support-services-netherlands)

Netcode Elektriciteit The codes are designed to give substance to existing laws and regulations and describe the way network operators behave among themselves and toward other connected parties:
<https://wetten.overheid.nl/BWBR0037940>

Imbalance pricing system
Details of the roles, responsibilities and methods applied in relation to the imbalance pricing system.
This document is available at [Documents general \(tennet.eu\)](https://tennet.eu/documents-general).

Implementation Guides
Many IT-related documents concerning the various communication channels with TenneT can be found on MyTenneT. This is a web portal for BSPs which is made accessible after requesting prequalification.

Historical prices and volumes:
Entso-e transparency platform: <https://transparency.entsoe.eu/>
average prices and contracted volumes incident reserves capacity bids

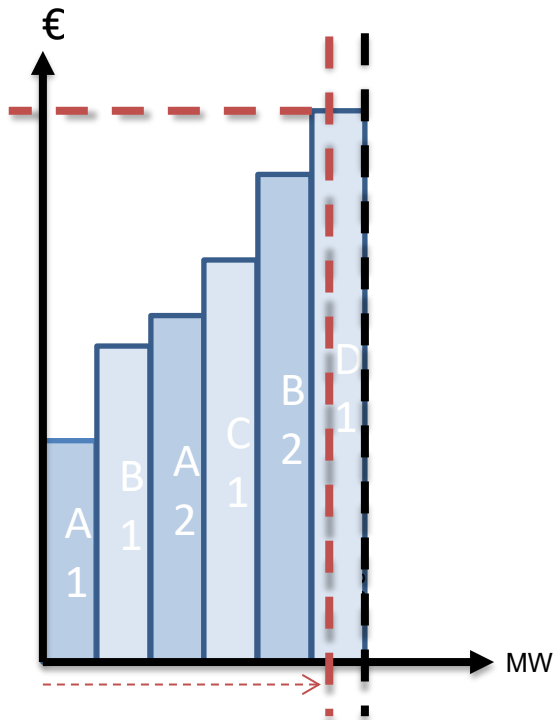
Balance Delta:
<https://www.tennet.eu/markets/transparency-data-netherlands/balance-delta>

Definitions

Terms used in this document which are not explained in the document are described in the "Begrippencode elektriciteit", <https://wetten.overheid.nl/BWBR0037938/2024-08-27>

12. Appendix 1 Activation scenarios

Initial activation



Example scenario:

ISP 1

Total MOL = 240 MW

6 bids of 40 MW each

4 BSP's (A, B, C, D)

IR need = 210 MW

Activated bids/BSP's in ISP 1:

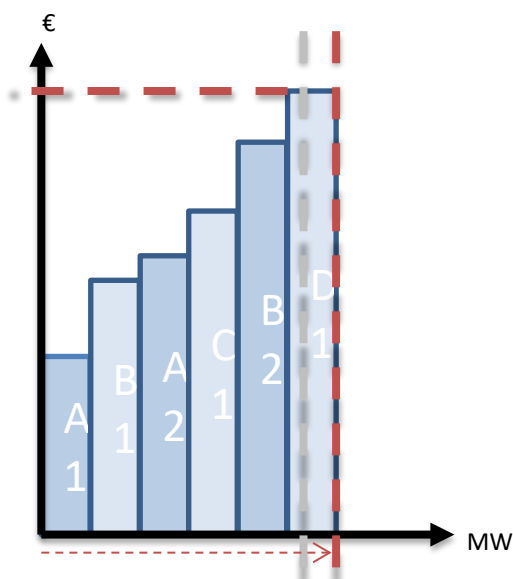
BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 10 MW (activation contains 1 partial bid)

Activation additional need



Example scenario:

Total MOL for ISP 1 and ISP 2 = 240 MW

6 bids of 40 MW each

4 BSP's (A, B, C, D)

IR need = 210 –increased to **240** MW

Activated bids/BSP's in ISP 1:

BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 10 MW (activation contains 1 partial bid)

Activated bids/BSP's in ISP 2:

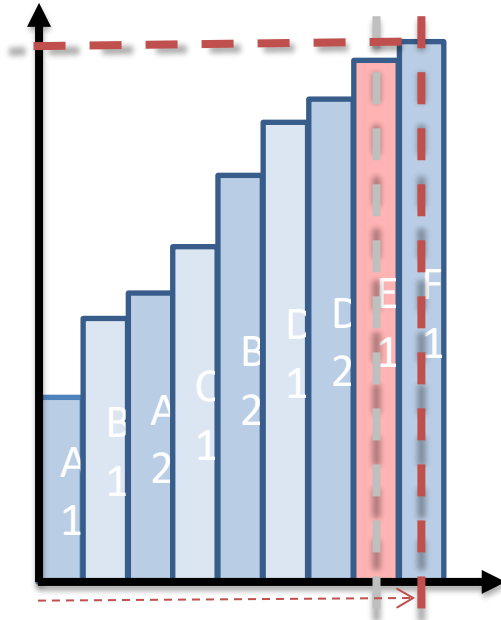
BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 40 MW (activation contains 1 bid)

Additional need with indivisible bids (1)



Example scenario:

Total MOL = 360 MW

9 bids of 40 MW each

6 BSP's (A, B, C, D, E, F)

The bid from BSP E is **not** divisible.

IR need = 210 –increased to **300** MW

Activated bids/BSP's ISP 1:

BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 10 MW (activation contains 1 partial bid)

Activated bids/BSP's in ISP 2:

BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

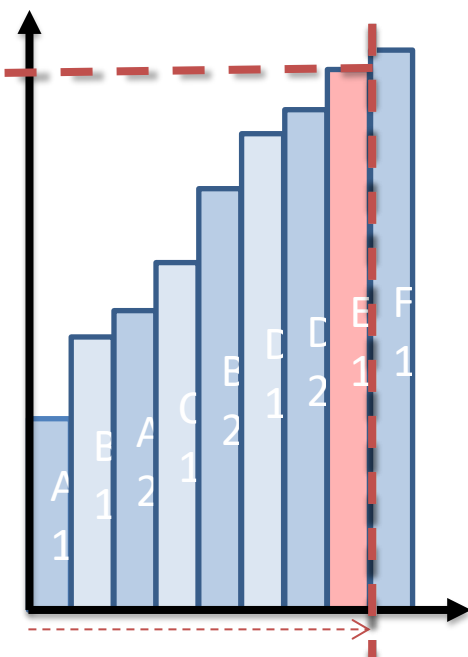
BSP C: 40 MW (activation contains 1 bid)

BSP D: 80 MW (activation contains 2 bids)

BSP E: skipped

BSP F: 20 MW (activation contains 1 partial bid)

Additional need with indivisible bids (2)



Example scenario:

Total MOL = 360 MW

9 bids of 40 MW each

6 BSP's (A, B, C, D, E, F)

The bid from BSP E is **not** divisible.

IR need = 210 increased to **320** MW

Activated bids/BSP's ISP 1:

BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 10 MW (activation contains 1 partial bid)

Activated bids/BSP's in ISP 2:

BSP A: 80 MW (activation contains 2 bids)

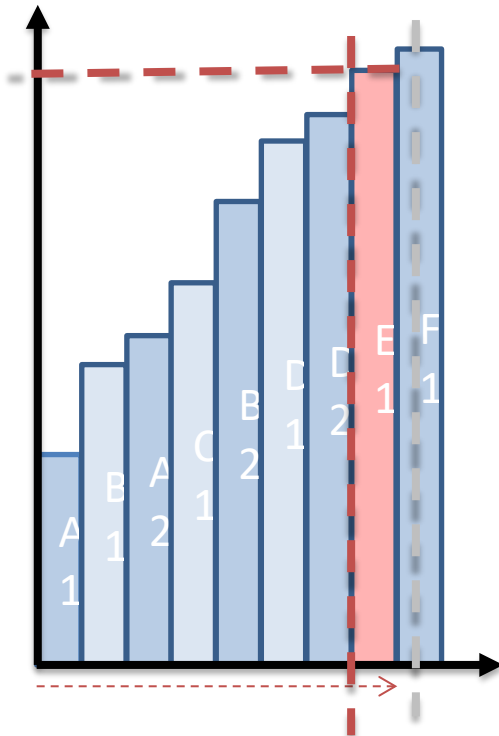
BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 80 MW (activation contains 2 bids)

BSP E: 40 MW (activation contains 1 bid)

Decreasing need with indivisible bids (1)



Example scenario:

Total MOL = 360 MW

9 bids of 40 MW each

6 BSP's (A, B, C, D, E, F)

The bid from BSP E is **not** divisible.

IR need = 320 decreased to **280** MW

Activated bids ISP 1:

BSP A: 80 MW (activation contains 2 bids)

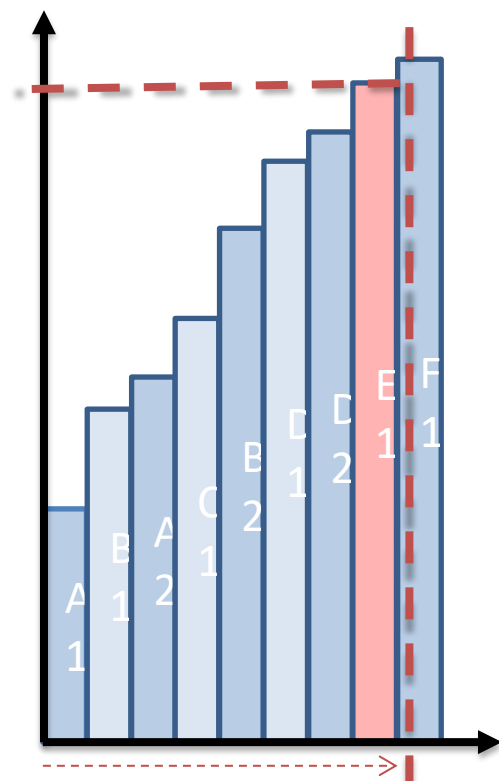
BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 80 MW (activation contains 2 bids)

BSP E: 40 MW (activation contains 1 bid) → **deactivated**

Decreasing need with indivisible bids (2)



Example scenario:

Total MOL = 360 MW

9 bids of 40 MW each

6 BSP's (A, B, C, D, E, F)

The bid from BSP E is **not** divisible.

IR need = 300 MW decreased to **290** MW

Activated bids/BSP's ISP 1:

BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

BSP D: 80 MW (activation contains 2 bids)

BSP E: 40 MW (activation contains 1 bid)

Activated bids/BSP's ISP 2:

BSP A: 80 MW (activation contains 2 bids)

BSP B: 80 MW (activation contains 2 bids)

BSP C: 40 MW (activation contains 1 bid)

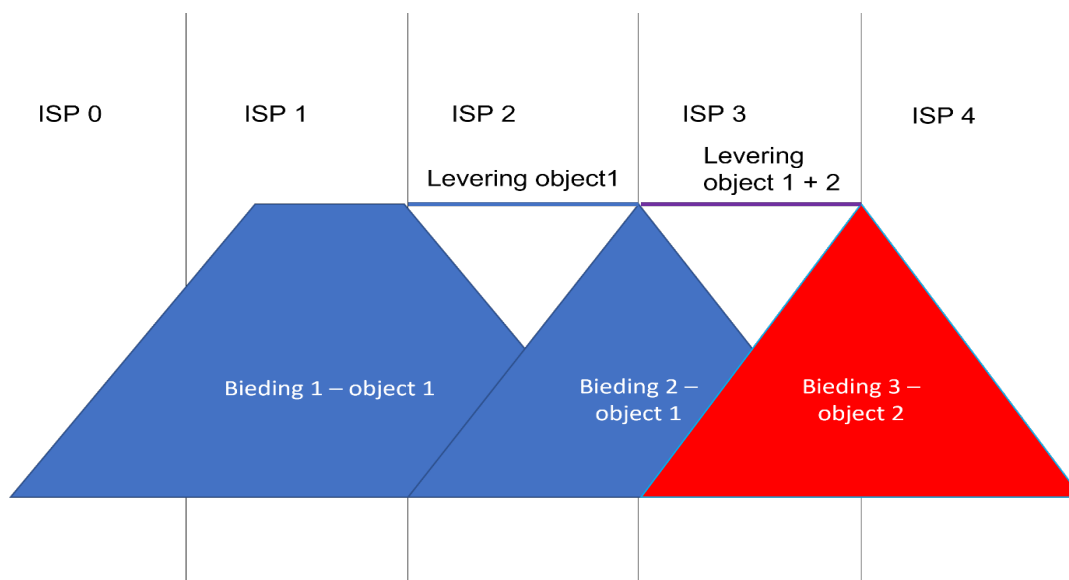
BSP D: 80 MW (activation contains 2 bids)

BSP E: skipped

BSP F: 10 MW (activation contains 1 bid)

13. Appendix 2 acceptable response per 5-minute periods

For the examples of the acceptable response the following activations will be used.



Bidding 2 is activated at T-0. For the activation of bidding 2 for 1 MW the expected energy is $1 \text{ MW} / 4 = 0,250 \text{ MWh}$. Based on the ramping 0,125 MWh will be delivered in ISP 2 and 0,125 MWh in ISP 3. The total expected minimal delivered energy in ISP 2 is $0,125 \text{ MWh} \times 90\% = 0,113 \text{ MWh}$.

After the preparation time of 5 minute in the 6^e minute, the minimum reaction is 0,001 MWh.

The expected energy delivered per 5-minute period:

Period	Delivered energy	Cumulated delivered energy
0 – 5 minutes	0 MWh	0 MWh
5- 10 minutes	More than 0,005 MWh	More than 0,005 MWh
10 – 15 minutes	Increased compared to the previous 5-minute period	Minimal 0,113 MWh

When bidding 1 is activated for ISP 1 at T-11 in ISP 0 (being the 4th minute of the first 5-minute measurement period of ISP 0):

ISP 0		
Period	Delivered energy	Cumulated delivered energy
0 – 5 minutes	0	0
5 – 10 minutes	0,001 MWh	0,001 MWh
10 – 15 minutes	More than 0,005 MWh	More than 0,006 MWh
ISP 1		
0 – 5 minutes	Increased compared to the previous 5-minute period	0,130 MWh being: 0,113 MWh + 0,017 MWh (1-minute stable delivery of 1MW)
5 – 10 minutes	0,083 MWh	0,213 MWh
10 – 15 minutes	0,083 MWh	0,296 MWh

When bidding 1 is activated for ISP 1 at T-7 in ISP 0 (being the 4th minute of the second 5-minute measurement period of ISP 0):

ISP 0		
Period	Delivered energy	Cumulated delivered energy
0 – 5 minutes	0	0
5 – 10 minutes	More than 0,002 MWh	More than 0,002 MWh
10 – 15 minutes	More than 0,005 MWh	More than 0,007 MWh
ISP 1		
0 – 5 minutes	Increased compared to the previous 5-minute period	Increasing towards 0,113 MWh
5 – 10 minutes	Increased compared to the previous 5-minute period	0,163 MWh being: 0,113 MWh + 3 x 0,017 MWh (1-minute stable delivery of 1MW)
10 – 15 minutes	0,083 MWh	0,246 MWh